SIAM J. MATH. ANAL.
Vol. 24, No. 6, pp. 1522–1536, November 1993
© 1993 Society for Industrial and Applied Mathematics
006

HIGHER-ORDER REGULARITY FOR THE SOLUTIONS OF SOME DEGENERATE QUASILINEAR ELLIPTIC EQUATIONS IN THE PLANE*

W. B. LIU† AND JOHN W. BARRETT†

Abstract. Local $C^{k,\beta}$ and $W^{2+k,\upsilon}$ ($k \geq 1, \beta > 0$, and $\upsilon \geq 1$) regularity is established for the solutions of a class of degenerate quasilinear elliptic equations, which include the p-Laplacian. Unlike the known local regularity results for such equations, k is larger than 2 in many notable cases. These results generalize those in [13], which were established only for the p-Laplacian. Furthermore, local results are extended to obtain a global regularity result in some cases. Global results of this type are essential in proving optimal error bounds for the finite element approximation of such equations.

Key words. regularity, degenerate, elliptic equations, quasilinear

AMS subject classifications. 35B65, 35J70, 35J60

1. Introduction.

Let Ω be a bounded open set in R^2 with a Lipschitz boundary $\partial\Omega$. Let φ be a positive continuous function on $(0, \infty)$. Many mathematical models from physical processes have the following form: given ρ, find u such that

$$-\nabla \cdot (\varphi(|\nabla u|)\nabla u) = \rho \quad \text{in } \Omega \tag{1.1}$$

plus some boundary conditions. Such models arise in fluid mechanics (see [4] and [5]), nonlinear diffusion (see [24]), and nonlinear elasticity (see [3]). An example of the latter is the mode III problem, longitudinal shear, for a power law material (see [3]). This leads to (1.1) with $\varphi(t) = t^{p-2}$ $(1 < p \leq 2)$, which is usually referred to as the p-Laplacian.

There is much work on the regularity of the solutions of (1.1) if $\varphi > 0$ is smooth on $[0, \infty)$ and satisfies certain coercivity and growth conditions. In this case the solutions of (1.1) will generally be smooth in any open set $G \subset\subset \Omega$. As key references, one can refer to [11], [12], [15], and [23]. These results are generally true for (1.1) when $\Omega \subset R^n$.

However, as seen above, in some physically relevant models φ may not be smooth on $[0, \infty)$ or may not satisfy the coercivity and growth conditions. Equations of this type are usually referred to as degenerate. A typical example is the p-Laplacian with $p \neq 2$. The well established regularity results for nondegenerate equations are generally not applicable to such cases (see [29] for some counterexamples). The regularity theory for these degenerate equations is much more complicated and until quite recently very little was known (see [31]).

Since the publication of [30], extensive work has been done on the regularity of the solutions of (1.1) and more general degenerate quasilinear equations and systems. One can find such work, for example, in [1], [2], [9], [10], [13], [17]–[19], [20]–[22], and [26]–[29]. Generally speaking, local and global $C^{1,\alpha}$ regularity has been established for the solutions of (1.1) and more general systems (for example, see [9], [18], [19], and [29]). In addition, some local H^2 regularity results have been established; for example, see [26] and [29]. These results remain true for (1.1) when $\Omega \subset R^n$. It should be noted that it is usually quite difficult to extend a local regularity result to a

* Received by the editors May 18, 1992; accepted for publication (in revised form) March 1, 1993.
† Department of Mathematics, Imperial College, London SW7 2BZ, United Kingdom. The work of the first author was supported by Science and Engineering Research Council of the United Kingdom grant GR/F81225.

Downloaded 09/18/13 to 128.101.152.17. Redistribution subject to SIAM license or copyright; see http://www.siam.org/journals/ojsa.php

Downloaded 09/18/13 to 128.101.152.17. Redistribution subject to SIAM license or copyright; see http://www.siam.org/journals/ojsa.php

global one due to the degenerate nonlinearity. In the plane there exist more powerful methods to study (1.1); for example, complex function theory. In the case $\varphi(t) = t^{p-2}$ and $\rho = 0$ a local $W^{2,2+\eta}(\Omega)$ regularity result, where $\eta(p) > 0$ has been proved, using the theory of quasi-regular mappings, firstly for $p \geq 2$ in [7] and then for $p > 1$ in [22]. In [20] global H^2 regularity results have been established for (1.1), for $\rho \in L^r(\Omega)$, $r > 2$, when $p \in (1, 2]$ and for $\rho = 0$ when $p > 2$, by using an a priori estimate for the solutions of a linear elliptic equation in the plane with only $L^\infty(\Omega)$ coefficients.

These global H^2 regularity results have been used in the error analysis of the finite element approximation of (1.1) in [6], [20], and [21]. From this work it is seen that the regularity of the solutions has a considerable influence on the order of convergence that can be proved for their finite element approximation. Taking the p-Laplacian ($1 < p < 2$) as an example, if $u \in W^{2,p}(\Omega)$ one can prove that the error for the continuous piecewise linear finite element approximation converges at the rate $h^{p/2}$ in $W^{1,p}(\Omega)$, whereas one can prove that it converges at the optimal rate h if $u \in C^{2,2/p-1}(\bar{\Omega}) \cap W^{3,1}(\Omega)$, see [6], or $u \in W^{3,p}(\Omega)$. With some restrictions on ρ and the boundary data, the global regularity $u \in W^{1+2/p,p}(\Omega)$ is shown in [21] to be sufficient to ensure the optimal convergence rate h. Thus one can see that it is important to establish some global higher-order regularity results in order to obtain optimal error bounds in $W^{1,p}(\Omega)$ for the continuous piecewise linear finite element approximation of (1.1).

A first step in this direction was taken in [13], where $C^{k,\alpha}$ and $W^{2+k,q}$ regularity has been established, using the theory of quasi-regular mappings for the solutions of the p-Laplacian in the plane with $\rho = 0$, although the results are only local. Unlike most known results, k is larger than 2 in many interesting cases; for example, the p-Laplacian with $1 < p \leq 2$. From this result it was pointed out in [21] that global $W^{3,p}(\Omega)$, $C^{2,2/p-1}(\bar{\Omega})$, and $W^{1+2/p,p}(\Omega)$ regularity can indeed be achieved for solutions of the p-Laplacian in some physically relevant cases, though this result is only applicable to a limited class of boundary data. It should be noted that the results in [13] are sharp and such high-order regularity (local or global) is not generally true for the case $\rho \neq 0$; see, for example, [17].

The purpose of this paper is to generalize the results in [13] to a class of degenerate quasilinear elliptic equations, which includes the p-Laplacian. Furthermore, we show that this local result can be extended to a global one in some cases.

Throughout this paper, we adopt the standard notation $W^{s,v}(\Omega)(s \geq 0,\ v \in [1, \infty])$ for Sobolev spaces on Ω with norm $\|\cdot\|_{W^{s,v}(\Omega)}$; see [14], for example. We denote $W^{m,2}(\Omega)$ by $H^m(\Omega)$ for nonnegative integers m and the spaces $W^{s,v}_{\text{loc}}(\Omega)$ consist of all functions which are in $W^{s,v}(D)$ for all open sets $D \subset\subset \Omega$ (that is, $\bar{D} \subset \Omega$). For the definitions of the spaces $C^{k,\beta}(\Omega)$ (k a nonnegative integer and $0 \leq \beta \leq 1$) and domains with $C^{k,\beta}$ boundaries; see [12], for example.

2. Preliminaries. Let Ω be a bounded open set in R^2 with a Lipschitz boundary $\partial\Omega$. Let φ satisfy the following conditions:

(A1) The function $t \to \varphi(t)t$ belongs to $C^0[0, \infty) \cap C^\infty(0, \infty)$.

(A2) There exist constants $q > 1$, $q \neq 2$, and $C_i, \kappa > 0$ such that for all $t \geq 0$

$$C_1 t \min\{t^{q-2}, (1+t)^{q-2}\} \leq \varphi(t)t \leq C_2 t \max\{t^{q-2}, (1+t)^{q-2}\}, \tag{2.1a}$$

and $\Gamma(t) \equiv \varphi'(t)t/\varphi(t)$ is such that for all $t \geq 0$,

$$\kappa \leq \Gamma(t)/(q-2) \leq 1. \tag{2.1b}$$

Downloaded 09/18/13 to 128.101.152.17. Redistribution subject to SIAM license or copyright; see http://www.siam.org/journals/ojsa.php

(A3) The function $\Gamma(t)$ is analytic at $t = 0$; that is, in an interval about $t = 0$,

$$\Gamma(t) \equiv \sum_{i=0}^{\infty} \gamma_i t^i. \tag{2.2a}$$

In addition we assume that there is a $\eta > 0$ such that

$$|\gamma_i| \leq |\gamma_0| \eta^i \quad \text{for all } i \geq 1, \tag{2.2b}$$

where we note from (2.1a) that $\Gamma(0) \equiv \gamma_0 \neq 0$.

We note that the regularity requirement $C^\infty(0, \infty)$ in (A1) can be appropriately relaxed and the results in this paper remain valid.

In this paper we will study the local and global regularity of the weak solutions $u \in W^{1,q}_{\text{loc}}(\Omega)(q > 1)$ of the following equation:

$$\nabla \cdot (\varphi(|\nabla u|)\nabla u) = 0 \quad \text{in } \Omega. \tag{2.3}$$

Example 2.1. For example, it is easy to see that $\varphi(t) \equiv t^{q-2}$ or $\varphi(t) \equiv [t(1 + t)]^{(q-2)/2}$ or $\varphi(t) \equiv [t(1 + t^2)]^{(q-2)/3}(q > 1)$ satisfies the conditions (A1)–(A3). It is also easy to show that if φ satisfies the conditions (A1) and (A2), then it satisfies the structural conditions (1.2)–(1.5) in [27] and (1.3)–(1.5) in [29].

Remark 2.1. We note that if $\varphi(t)$ satisfies the conditions (A1)–(A3) and u satisfies (2.3), then for any constant $\lambda > 0$, $\equiv \lambda^{-1}u$ is such that $\nabla \cdot (\hat{\varphi}(|\nabla \hat{u}|)\nabla \hat{u}) = 0$ in Ω with $\hat{\varphi}(t) \equiv \varphi(\lambda t)$. Therefore $\hat{\varphi}(t)$ satisfies (A1)–(A3) with $\hat{\Gamma}(t) \equiv \hat{\varphi}'(t)t/\hat{\varphi}(t) \equiv \Gamma(\lambda t)$, $\hat{\Gamma}(t) \equiv \sum_{i=0}^{\infty} \hat{\gamma}_i t^i$, and $|\hat{\gamma}_i| \leq |\hat{\gamma}_0|(\lambda\eta)^i$ for all $i \geq 1$. Therefore without loss of generality we can assume that $\eta > 0$ in (2.2b) is as small as we please throughout this paper.

In [8], [19], [27], and [29] the existence of solutions, $u \in W^{1,q}_{\text{loc}}(\Omega)$, to the weak formulation of (2.3) is established. It follows from [9] and [29] that $u \in C^{1,\alpha}(\Omega)$ for an $\alpha > 0$. Furthermore, if some suitable boundary conditions are imposed, the weak solution $u \in W^{1,q}(\Omega)$ is unique.

Let u be a weak solution of (2.3), and let $f \equiv u_x - iu_y$ be the complex gradient of u. It turns out that regularity of f, and hence u, can be more easily examined if we view it as a function of a complex variable. To this end we first express (2.3) in the form of a complex equation. Let us recall for $z \equiv x + iy$, $(x, y) \in R^2$ that $\partial/\partial z \equiv (\partial/\partial x - i\partial/\partial y)/2$ and $\partial/\partial \bar{z} \equiv (\partial/\partial x + i\partial/\partial y)/2$.

LEMMA 2.1. *The mapping $f \equiv u_x - iu_y$ is K-quasi-regular with $K = \max\{q - 1, 1/(q-1)\}$. Hence $f \in [W^{1,2+\eta}_{\text{loc}}(\Omega)]^2$ for a $\eta > 0$ and $f^{-1}(0) \equiv \{z \in \Omega : \nabla u(z) = 0\}$ is discrete provided that u is not identically constant.*

Let $p \equiv 2 + \Gamma(0)$; then

$$p \in [q, 2) \quad \text{if } q \in (1, 2) \quad \text{and} \quad p \in (2, q] \quad \text{if } q > 2. \tag{2.4}$$

On setting $G(t) \equiv \Gamma(t)/\Gamma(0)$ and $F(t) \equiv pG(t)/(2 + (p - 2)G(t))$, it follows that f satisfies for almost every $z \in \Omega$,

$$\partial f/\partial \bar{z} = (1/p - \tfrac{1}{2})[(\bar{f}/f)(\partial f/\partial z) + (f/\bar{f})(\overline{\partial f/\partial z})]F(|f|). \tag{2.5}$$

Proof. We will show that f is a quasi-regular mapping by generalizing the approach given in [22] for the p-Laplacian. Firstly we prove that $u \in H^2_{\text{loc}}(\Omega)$.

Downloaded 09/18/13 to 128.101.152.17. Redistribution subject to SIAM license or copyright; see http://www.siam.org/journals/ojsa.php

Let $\varphi^\varepsilon(t) \equiv \varphi((t^2+\varepsilon)^{1/2})$ and u^ε be the unique solution of the following problem:

$$\nabla \cdot (\varphi^\varepsilon(|\nabla u^\varepsilon|)\nabla u^\varepsilon) = 0 \quad \text{in } B \quad \text{and} \quad u^\varepsilon = u \quad \text{on } \partial B, \tag{2.6}$$

where B is an open ball in Ω. As $u \in C^1(\bar{B})$, it follows from [12] and [15] that $u^\varepsilon \in H^2_{\text{loc}}(B) \cap C^1(B)$ (actually $u^\varepsilon \in C^{2,\beta}(B)$ for a $\beta > 0$). Noting that our Assumptions (A1) and (A2) are equivalent to (1.2)–(1.5) in [27], it follows that $u_\varepsilon \to u$ in $W^{1,q}(B)$ as $\varepsilon \to 0$. On the other hand, (2.6) can be restated as

$$a^\epsilon \partial^2 u^\epsilon/\partial x^2 + 2b^\epsilon \partial^2 u^\epsilon/\partial x \partial y + c^\epsilon \partial^2 u^\epsilon/\partial y^2 = 0 \quad \text{in } B \quad \text{and} \quad u^\epsilon = u \quad \text{on } \partial B, \tag{2.7}$$

where $v \equiv (\epsilon + |\nabla u^\epsilon|^2)^{1/2}, v_1 \equiv \partial u^\epsilon/\partial x, v_2 \equiv \partial u^\epsilon/\partial y, a^\epsilon \equiv [1 + [\varphi'(v)v/\varphi(v)](v_1/v)^2]$, $b^\epsilon \equiv [\varphi'(v)v/\varphi(v)](v_1 v_2/v^2)$, and $c^\epsilon \equiv [1 + [\varphi'(v)v/\varphi(v)](v_2/v)^2]$. Setting $d^\epsilon \equiv a^\epsilon c^\epsilon - (b^\epsilon)^2 \equiv 1 + [\varphi'(v)v/\varphi(v)][\{(v_1)^2 + (v_2)^2\}/v^2]$, we have from (2.1b) that

$$\min\{1, q-1\} \le a^\epsilon, c^\epsilon, d^\epsilon \le \max\{1, q-1\} \quad \text{and} \quad |b^\epsilon| \le |q-2|; \tag{2.8}$$

that is, the equation is uniformly (with respect to ϵ) elliptic with bounded coefficients.

We now apply the inequality (17.20) in [15, p. 228]. For any $B'' \subset\subset B' \subset\subset B$ this yields that $|u^\epsilon|_{H^2(B'')} \le C\|u^\epsilon\|_{H^1(B')}$, where C is independent of ϵ. In [27] it has been shown that $|u^\epsilon| + |\nabla u^\epsilon|$ is bounded on B' as $\epsilon \to 0$ since $u \in C^0(\bar{B})$. This implies that $\|u^\epsilon\|_{H^2(B'')}$ is uniformly bounded. Hence $u \in H^2(B'')$ so that $u \in H^2_{\text{loc}}(\Omega)$, since u^ϵ will converge to u weakly in $H^2(B'')$ as $\epsilon \to 0$. Since the proof of the uniform boundedness of $|u^\epsilon| + |\nabla u^\epsilon|$ in [27] is quite complicated, we now give a simpler proof. Let $u_0 = \min_{x \in \partial B} u(x)$ and $u_1 = \max_{x \in \partial B} u(x)$. Note that u_0 and u_1 are constants so that one can apply Lemma 3 in [27] to (2.6). This shows that $u_0 \le u^\epsilon \le u_1$ for any $\epsilon > 0$. Then by applying Theorem 12.4 in [12] with $f = 0$ and the interpolation inequality (12.23)[(6.82)] in [12] one can see that $|u^\epsilon| + |\nabla u^\epsilon|$ is bounded on B' as $\epsilon \to 0$. Of course for $q > 2$, $\|u^\epsilon\|_{H^1(B)}$ is uniformly bounded since $\|u^\epsilon\|_{W^{1,q}(B)}$ is. Therefore, we have that $u \in H^2_{\text{loc}}(\Omega)$, which implies that $f \in [H^1_{\text{loc}}(\Omega)]^2$.

Let $f^\epsilon \equiv (v_1, -v_2)$. Adopting the same notation as in Theorem 1 in [22] we have from (2.8) that for all $z \in B$,

(2.9)

$$\begin{aligned}
&\|Df^\epsilon(z)\|^2/Jf^\epsilon(z) \\
&\quad \equiv [(\partial v_1/\partial x)^2 + (\partial v_1/\partial y)^2 + 2(\partial v_1/y)^2](z)/[(\partial v_1/\partial y)^2 - (\partial v_1/\partial x)(\partial v_2/\partial y)](z) \\
&\quad \le [(a^\epsilon)^2 + 2(b^\epsilon)^2 + (c^\epsilon)^2](z)/[a^\epsilon c^\epsilon - (b^\epsilon)^2](z) \\
&\quad = [1 + (d^\epsilon)^2](z)/d^\epsilon(z) \le 2K,
\end{aligned}$$

where $K = \max\{q-1, 1/(q-1)\}$. Hence f^ϵ is a K-quasi-regular mapping; see [13] and [22]. As $|u^\epsilon| + |\nabla u^\epsilon|$ is bounded on B' as $\epsilon \to 0$, we can apply the argument in Theorem 1 in [22] and let $\epsilon \to 0$ in the above to yield that f is K-quasi-regular with $K = \max\{q-1, 1/(q-1)\}$.

It follows immediately from the theory of quasi-regular maps, see [15], that $f \in [W^{1,2+\eta}_{\text{loc}}(\Omega)]^2$ for a $\eta > 0$ and $f^{-1}(0) \equiv \{z \in \Omega : \nabla u(z) = 0\}$ is discrete, provided that u is not identically constant.

The result (2.4) follows immediately from Assumption (A2), which in turn implies that G and F are well defined. Finally we note that if $|\nabla u(z)| > 0$, then (2.3) yields

Downloaded 09/18/13 to 128.101.152.17. Redistribution subject to SIAM license or copyright; see http://www.siam.org/journals/ojsa.php

that

(2.10)
$$2\partial f(z)/\partial\bar{z} \equiv \Delta u(z) = -\Gamma(|\nabla u(z)|)[(u_x)^2u_{xx} + 2u_xu_yu_{xy} + (u_y)^2u_{yy}](z)/|\nabla u(z)|^2$$
$$= (2-p)G(|\nabla u(z)|)[(u_x)^2u_{xx} + 2u_xu_yu_{xy} + (u_y)^2u_{yy}](z)/|\nabla u(z)|^2.$$

In addition, if $|\nabla u(z)| > 0$, then $2\partial f/\partial z \equiv u_{xx} - u_{yy} - 2iu_{xy}$, and so

(2.11)
$$[(\bar{f}/f)(\partial f/\partial z) + (f/\bar{f})(\overline{\partial f/\partial z})](z)$$
$$\equiv -\Delta u(z) + 2[(u_x)^2u_{xx} + 2u_xu_yu_{xy} + (u_y)^2u_{yy}](z)/|\nabla u(z)|^2.$$

Combining (2.10), (2.11), and noting that $\{z \in \Omega : \nabla u(z) = 0\}$ is discrete yields the desired result (2.5). □

We note that in the case $\varphi(t) = t^{q-2}, q \in (1,\infty)$, it was first proved in [7] for $q > 2$ and then in [22] for $q > 1$ that the corresponding map f was quasi regular.

It follows from classical regularity theory that the solution u of (2.3), and the corresponding f, is C^∞ outside the set $\{z \in \Omega : \nabla u(z) = 0\}$ (see, for example, [12]). Therefore it is sufficiently general to examine the regularity of u (or f) near one of the zero points of f, which one may assume is the origin. From now on, we will assume that $0 \in \Omega$ and $f(0) = 0$.

We now consider the regularity of local solutions of (2.5). We generalize the approach for the p-Laplacian given in [13]. From the theory of quasi-regular mappings, see [16], and as $f(0) = 0$ with $0 \in \Omega$ is an isolated zero of f, there exists a quasi-conformal homeomorphism χ defined in a neighborhood of $z = 0$ and a positive integer n such that $\chi(0) = 0$ and

(2.12)
$$f(z) \equiv [\chi(z)]^n.$$

From (2.5) and (2.12) we have that

$$\partial\chi/\partial\bar{z} = (1/p - \tfrac{1}{2})[(\bar{\chi}^n/\chi^n)(\partial\chi/\partial z) + (\chi/\bar{\chi})(\overline{\partial\chi/\partial z})]F(|\chi|^n).$$

Let $H(\xi)$ be the inverse of $\chi(z)$ in a neighborhood of $z = 0$. It follows that H is quasi-conformal in a neighborhood B of $\xi = 0$, and hence $H \in C(B) \cap H^1(B)$ and $H \in C^\infty(B - \{0\})$. Then we have that

(2.13)
$$\partial H/\partial\bar{\xi} = (\tfrac{1}{2} - 1/p)[(\bar{\xi}^n/\xi^n)(\overline{\partial H/\partial\xi}) + (\xi/\bar{\xi})(\partial H/\partial\xi)]F(|\xi|^n).$$

Let $\xi = re^{i\theta}$ and $H(r,\theta) \equiv H(re^{i\theta})$. Proceeding as in [13] one obtains that

(2.14)
$$2rH_r = -i(2 + (p-2)G(r^n))H_\theta + i(p-2)G(r^n)e^{-2in\theta}\overline{H_\theta}.$$

Expanding $H(r,\theta)$ into Fourier series with respect to θ, $0 \le \theta < 2\pi$,

(2.15)
$$H(r,\theta) \equiv \sum_{k=-\infty}^{\infty} h_k(r)e^{i(k-n)\theta},$$

where $h_k \in C[0,\delta] \cap C^\infty(0,\delta]$ for $\delta > 0$. It follows from (2.14) and (2.15) that for all integers k,

(2.16a)
$$2rh_k'(r) = -(n-k)(2 + (p-2)G(r^n))h_k(r) - (n+k)(p-2)G(r^n)\overline{h_{-k}(r)}$$

Downloaded 09/18/13 to 128.101.152.17. Redistribution subject to SIAM license or copyright; see http://www.siam.org/journals/ojsa.php

and

$$2r\overline{h'_{-k}(r)} = -(n+k)(2+(p-2)G(r^n))\overline{h_{-k}(r)} - (n-k)(p-2)G(r^n)h_k(r). \tag{2.16b}$$

Setting $X_k(r) \equiv (h_k(r), \overline{h_{-k}(r)})^T$, this system of equations can be restated as follows: Let n be a positive integer and $p \in (1,2) \cup (2,\infty)$. Then for any integer k find $X_k(r)$ such that

$$2rX'_k(r) = -[A(k) + \hat{g}(r)B(k)]X_k(r), \tag{2.17a}$$

where

(2.17b)

$$A(k) \equiv \begin{pmatrix} (n-k)p & (n+k)(p-2) \\ (n-k)(p-2) & (n+k)p \end{pmatrix}, \qquad B(k) \equiv \begin{pmatrix} (n-k) & (n+k) \\ (n-k) & (n+k) \end{pmatrix},$$

and

$$\hat{g}(r) \equiv (p-2)(G(r^n)-1). \tag{2.17c}$$

The advantage of this reformulation is that the structure of the general solutions for the system $2rW'_k(r) = -A(k)W_k(r)$ is easily established when $k \neq -n$ and the function $\hat{g}(r)B(k)/r$ is regular.

Therefore, to study the regularity of local solutions of (2.5) we need to study the structure of the general solutions about $r = 0$ to the system (2.17). This we do in the next section.

3. Series expansion for $H(\xi)$. We will analyze the system (2.17) by developing a series solution about $r = 0$. In fact, we analyze a more general system: (2.17a, b) with $\hat{g}(r)$ being any real analytical function at $r = 0$ satisfying $\hat{g}(0) = 0$, instead of the specific choice (2.17c). First, we need a lemma.

LEMMA 3.1. *Let m be a positive integer and let $M \in R$ be such that $M \geq 1+2^m$. Then for any integer $k \in [0,m]$ and any integer $i \geq 1$,*

$$\sum_{j=0}^{i-1}(i-j)^k M^j \leq M^i. \tag{3.1}$$

Proof. Simple calculations yield that (3.1) is true for (i) $k = 0$ for any integer $i \geq 1$, and (ii) $i = 1$ for any integer $k \in [0,m]$. Now assume (3.1) holds for all $k \in [0,\hat{m}]$ and any integer $i \geq 1$ with $0 \leq \hat{m} < m$. Suppose that for $k = \hat{m}+1$ it is also true for any integer $i \in [1,\hat{i}]$ with $\hat{i} \geq 1$. Then we have that

$$\begin{aligned}
\sum_{j=0}^{\hat{i}}(\hat{i}+1-j)^{\hat{m}+1}M^j &= (\hat{m}+1)!\sum_{j=0}^{\hat{i}}M^j\sum_{k=0}^{\hat{m}+1}(\hat{i}-j)^k \Big/ [k!(\hat{m}+1-k)!] \\
&= M^{\hat{i}} + \sum_{k=0}^{\hat{m}+1}\{(\hat{m}+1)!/[k!(\hat{m}+1-k)!]\}\sum_{j=0}^{\hat{i}-1}(\hat{i}-j)^k M^j \\
&\leq \left[1+\sum_{k=0}^{\hat{m}+1}\{(\hat{m}+1)!/[k!(\hat{m}+1-k)!]\}\right]M^{\hat{i}} \\
&\leq (1+2^{\hat{m}+1})M^{\hat{i}} \leq (1+2^m)M^{\hat{i}} \leq M^{\hat{i}+1}.
\end{aligned}$$

Downloaded 09/18/13 to 128.101.152.17. Redistribution subject to SIAM license or copyright; see http://www.siam.org/journals/ojsa.php

Therefore, (3.1) also holds for $k = \hat{m} + 1$ and $i = \hat{i} + 1$. Hence by induction (3.1) holds for any integer $k \in [0, m]$ and any integer $i \geq 1$. □

THEOREM 3.1. *Let $p \in (1,2) \cup (2,\infty)$ and n be a positive integer. Suppose that $\hat{g}(r) \equiv \sum_{i=1}^{\infty} g_i r^i$ near $r = 0$ and that there exist $\mathcal{G} \in R^+$ and a positive integer m such that $|g_i| \leq \mathcal{G} i^m$ for all $i \geq 1$. Then for any integer k, except $k = -n$, the general solution $X_k(r)$ of* (2.17 a, b) *near $r = 0$ is of the form*

$$X_k(r) \equiv C_k^+ r^{\lambda_k^+} X_k^+(r) + C_k^- r^{\lambda_k^-} [X_k^-(r) + cr^{\rho_k} \ln(r) X_k^+(r)], \tag{3.2a}$$

where $C_k^{\pm}$ are general constants,

$$\lambda_k^{\pm} \equiv \{-pn \pm [4k^2(p-1) + n^2(p-2)^2]^{1/2}\}/2, \tag{3.2b}$$

$$\rho_k \equiv (\lambda_k^+ - \lambda_k^-) \equiv [4k^2(p-1) + n^2(p-2)^2]^{1/2}, \tag{3.2c}$$

$$\epsilon_k^{\pm} \equiv [(n-k)p + 2\lambda_k^{\pm}]/[(n+k)(2-p)]. \tag{3.2d}$$

The constant c is zero if ρ_k is not an integer, but possibly nonzero otherwise,

$$X_k^{\pm}(r) \equiv \begin{pmatrix} \hat{a}_k^{\pm}(r) + \hat{b}_k^{\pm}(r) \\ \epsilon_k^+ \hat{a}_k^{\pm}(r) + \epsilon_k^- \hat{b}_k^{\pm}(r) \end{pmatrix}, \tag{3.2e}$$

where $\hat{a}_k^{\pm}(0) + \hat{b}_k^{\pm}(0) = 1$ and

$$\hat{a}_k^{\pm}(r) \equiv \sum_{i=0}^{\infty} a_i^{\pm}(k) r^i, \qquad \hat{b}_k^{\pm}(r) \equiv \sum_{i=0}^{\infty} b_i^{\pm}(k) r^i. \tag{3.2f}$$

In addition there exist constants $C(k)$ and $M \geq 1 + 2^m$, dependent on p and n such that $|a_i^{\pm}(k)|, |b_i^{\pm}(k)| \leq C(k) M^i$ for all $i \geq 0$.

Proof. Let $\mathcal{X}(r)$ be a fundamental solution to the system $2rW_k'(r) = -A(k)W_k(r)$ when $k \neq n$; that is,

$$\mathcal{X}_k(r) \equiv \begin{pmatrix} r^{\lambda_k^+} & r^{\lambda_k^-} \\ \epsilon_k^+ r^{\lambda_k^+} & \epsilon_k^- r^{\lambda_k^-} \end{pmatrix},$$

where $\lambda_k^{\pm}$ are the eigenvalues of $-A(k)/2$ and $(1, \epsilon_k^{\pm})^T$ the corresponding eigenvectors; see (3.2b, d). We now seek two linearly independent solutions of (2.17a, b) of the form $\mathcal{X}_k(r) Y_k(r)$. Inserting $\mathcal{X}_k(r) Y_k(r)$ into (2.17a) one obtains

$$2rY_k'(r) = -\hat{g}(r) \mathcal{X}_k^{-1}(r) B(k) \mathcal{X}_k(r) Y_k(r)$$

and hence

$$Y_k'(r) = [\hat{g}(r)/r] D_k(r) Y_k(r), \tag{3.3}$$

Downloaded 09/18/13 to 128.101.152.17. Redistribution subject to SIAM license or copyright; see http://www.siam.org/journals/ojsa.php

where

$$D_k(r) \equiv \begin{pmatrix} d_1(k) & d_2(k)r^{-\rho_k} \\ d_3(k)r^{\rho_k} & d_4(k) \end{pmatrix}$$

$$\equiv (\epsilon_k^- - \epsilon_k^+)^{-1} \begin{pmatrix} (\epsilon_k^- - 1)\tau_k^+ & (\epsilon_k^- - 1)\tau_k^- r^{-\rho_k} \\ (1-\epsilon_k^+)\tau_k^+ r^{\rho_k} & (1-\epsilon_k^+)\tau_k^- \end{pmatrix}$$

with

$$\tau_k^\pm \equiv (n-k) + \epsilon_k^\pm (n+k).$$

It follows that there exists a $\mathcal{D} > 1$, independent of p, n, and k, such that

$$|d_j(k)| \le \mathcal{D}(|k| + n), \qquad j = 1 \to 4. \tag{3.4}$$

From our assumptions we have that $\hat{g}(r)/r \equiv \sum_{i=0}^{\infty} g_{i+1} r^i$. We now look for two linearly independent solutions $Y_k^\pm$ to (3.3) of the form

$$Y_k^\pm(r) \equiv \begin{pmatrix} r^{\alpha_k^\pm}\ \hat{a}_k^\pm(r) \\ r^{\beta_k^\pm}\ \hat{b}_k^\pm(r) \end{pmatrix} \equiv \begin{pmatrix} \sum_{i=0}^{\infty} a_i^\pm(k) r^{\alpha_k^\pm + i} \\ \sum_{i=0}^{\infty} b_i^\pm(k) r^{\beta_k^\pm + i} \end{pmatrix}. \tag{3.5}$$

Inserting (3.5) into (3.3) and equating coefficients one obtains the following linearly independent solutions:

$Y_k^+(r)$ by choosing $\alpha_k^+ \equiv 0, \beta_k^+ \equiv \rho_k, a_0^+(k) \equiv 1, b_0^+(k) \equiv 0$ and for $i \ge 0$,

$$\begin{aligned} a_{i+1}^+(k) &\equiv \left[\sum_{j=0}^{i} g_{i+1-j}(d_1(k)a_j^+(k) + d_2(k)b_j^+(k))\right] \Big/ (i+1), \\ b_{i+1}^+(k) &\equiv \left[\sum_{j=0}^{i} g_{i+1-j}(d_3(k)a_j^+(k) + d_4(k)b_j^+(k))\right] \Big/ (i+1+\rho_k); \end{aligned} \tag{3.6a}$$

if ρ_k is not an integer:

$Y_k^-(r)$ by choosing $\alpha_k^- \equiv -\rho_k$, $\beta_k^- \equiv 0$, $a_0^-(k) \equiv 0$, $b_0^-(k) \equiv 1$ and for $i \ge 0$,

$$\begin{aligned} a_{i+1}^-(k) &\equiv \left[\sum_{j=0}^{i} g_{i+1-j}(d_1(k)a_j^-(k) + d_2(k)b_j^-(k))\right] \Big/ (i+1-\rho_k), \\ b_{i+1}^-(k) &\equiv \left[\sum_{j=0}^{i} g_{i+1-j}(d_3(k)a_j^-(k) + d_4(k)b_j^-(k))\right] \Big/ (i+1). \end{aligned} \tag{3.6b}$$

Then we obtain two linearly independent solutions $r^{\lambda_k^+} X_k^+(r)$ and $r^{\lambda_k^-} X_k^-(r)$ to (2.17a, b), where $r^{\lambda_k^\pm} X_k^\pm(r) \equiv \mathcal{X}_k(r) Y_k^\pm(r)$. Hence the general solution is given by (3.2a)

with $c = 0$ and $X_k^{\pm}(r)$ defined by (3.2e) with $\hat{a}_k^{\pm}(0) + \hat{b}_k^{\pm}(0) \equiv 1$. Furthermore, if $C(k)$ and M are such that $M \geq 1 + 2^m$ and $|a_i^{\pm}(k)|$, $|b_i^{\pm}(k)| \leq C(k)M^i$ for all $i < 2\mathcal{GD}(|k| + n) + \rho_k$ it follows from (3.6), (3.5), and Lemma 3.1 that $|a_i^{\pm}(k)|$, $|b_i^{\pm}(k)| \leq C(k)M^i$ for all $i \leq 0$.

If ρ_k is an integer, say $l \geq 1$, then the solution (3.6b) is no longer valid. Taking $Y_k^+(r)$ as in (3.6a) to be one solution to (3.3), a routine calculation yields that a second solution, linearly independent to $Y_k^+(r)$, is $Y_k^-(r) + c\ln(r)Y_k^+(r)$, where

$$c \equiv \sum_{j=0}^{l-1} g_{l-j}(d_1(k)a_j^+(k) + d_2(k)b_j^+(k))$$

and $Y_k^-(r)$ is given by (3.5) with $\alpha_k^- \equiv -\rho_k \equiv -l$, $\beta_k^- \equiv 0$, $a_0^-(k) \equiv 0$, $b_0^-(k) \equiv 1$. For $i = 0 \to l-2$, $a_{i+1}^-(k)$ is defined by (3.6b), $a_{l-1}^-(k) \equiv 0$, for $i = 0 \to l-1$, $b_{i+1}^-(k)$ is defined by (3.6b), and finally for $i \geq l$,

(3.7)

$$a_{i+1}^-(k) \equiv \left[-ca_{i+1-l}^+(k) + \sum_{j=0}^{i} g_{i+1-j}(d_1(k)a_j^-(k) + d_2(k)b_j^-(k))\right] \Big/ (i+1-l),$$

$$b_{i+1}^-(k) \equiv \left[-cb_{i+1-l}^+(k) + \sum_{j=0}^{i} g_{i+1-j}(d_3(k)a_j^-(k) + d_4(k)b_j^-(k))\right] \Big/ (i+1).$$

We obtain two linearly independent solutions $r^{\lambda_k^+}X_k^+(r)$ and $r^{\lambda_k^-}[X_k^-(r) + cr^{\rho_k}\ln(r)X_k^+(r)]$ to (2.17 a, b) with c possibly nonzero, by setting $r^{\lambda_k^{\pm}}X_k^{\pm}(r) \equiv \mathcal{X}_k(r)Y_k^{\pm}(r)$. It follows that $X_k^{\pm}(r)$ are given by (3.2e) with $\hat{a}_k^{\pm}(0) + \hat{b}_k^{\pm}(0) \equiv 1$. Similarly to the above we can choose constants $C(k)$ and $M \geq 1 + 2^m$, dependent on p and n, such that $|a_i^{\pm}(k)|$, $|b_i^{\pm}(k)| \leq C(k)M^i$ for all $i \geq 0$. □

With the help of Lemma 3.1 and the following corollary to Theorem 3.1 we will establish our main regularity result for (2.3) in the next section.

COROLLARY 3.1. *Let $p \in (1,2) \cup (2,\infty)$ and n be a positive integer. Suppose that $\hat{g}(r) \equiv \sum_{i=1}^{\infty} g_i r^i$ near $r = 0$ and that $|g_i| \leq \eta^i$ for all $i \geq 1$, where $\eta \leq 1$. Then for any integer k, except $k = -n$, the general solution $X_k(r)$ of* (2.17 a, b) *near $r = 0$ is of the form* (3.2). *In addition for $k \geq n$ and for $\eta \leq 1/(8\mathcal{D})$ we have that*

$$|a_i^+(k)|, |b_i^+(k)| \leq (8\mathcal{D}\eta k)^i/i!, \qquad i = 0 \to k, \tag{3.8a}$$

$$|a_i^+(k)|, |b_i^+(k)| \geq (8\mathcal{D}\eta k)^k/k!, \qquad i \geq k, \tag{3.8b}$$

where $\mathcal{D} > 1$ is the constant, independent of p, n, and k, appearing in (3.4).

Proof. Clearly Theorem 3.1 applies with the given assumptions on $\hat{g}$. Let $M \equiv 8\mathcal{D}\eta k$. Firstly, we note that for any integer $j \in [1,k]$,

$$I_j \equiv \sum_{i=0}^{j-1} \eta^{j-i}M^i/i! \leq M^j/j!. \tag{3.9}$$

Equation (3.9) holds for $j = 1$ as $\eta \leq 1$. Assume it is true for $1 \leq j \leq l < k$, then $I_{l+1} \equiv \eta(I_l + M^l/l!) \leq 2\eta M^l/l! \leq M^{l+1}/(l+1)!$. Hence, by induction (3.9) holds for $1 \leq j \leq k$.

Downloaded 09/18/13 to 128.101.152.17. Redistribution subject to SIAM license or copyright; see http://www.siam.org/journals/ojsa.php

Downloaded 09/18/13 to 128.101.152.17. Redistribution subject to SIAM license or copyright; see http://www.siam.org/journals/ojsa.php

We now show that the improved bounds on the coefficients, (3.8), hold. (3.8a) holds for $i = 0$. Assume it is true for $0 \le i \le l < k$; then from (3.6a), (3.4), and (3.9) we have that

$$|a_{l+1}^{+}(k)|, |b_{l+1}^{+}(k)| \le 4\mathcal{D}k \left[\sum_{j=0}^{l} \eta^{l+1-j} M^j/j!\right] \Big/ (l+1)$$

$$\le M\left[I_{l-1} + M^l/1!\right]/2(l+1) \le M^{l+1}/(l+1)!.$$

Hence, by induction (3.8a) holds. Inequality (3.8b) holds for $i = k$. Assume it is true for $k \le i \le l$. Then as $M \le k$ we have from (3.6a), (3.4), (3.9), and (3.8a) that

$$|a_{l+1}^{+}(k)|, |b_{l+1}^{+}(k)| \le [8\mathcal{D}\eta k/(l+1)] \left[\sum_{j=0}^{k-1} \eta^{l-j} M^j/j! + M^k/k! \sum_{j=k}^{l} \eta^{l-j}\right] \Big/ 2$$

$$\le \left[\eta^{l-k} + \sum_{j=0}^{l-k} \eta^j\right] (M^k/k!)/2 \le (M^k/k!).$$

Hence, by induction (3.8b) holds. □

4. The main regularity result.

THEOREM 4.1. *Let $u \in W_{\text{loc}}^{1,q}(\Omega)$, $1 < q < \infty$, be a local weak solution of* (2.3) *with φ satisfying* (A1)–(A3). *Then*

$$u \in C^{k,\beta}(\Omega) \cap W_{\text{loc}}^{k+2,\upsilon}(\Omega), \tag{4.1}$$

where the integer $k \ge 1$ and exponents $\beta \in (0,1]$ and $\upsilon \in [1,2]$ are determined uniquely from

$$6(k+\beta) = 7 + 1/(p-1) + [1 + 14/(p-1) + 1/(p-1)^2]^{1/2}$$
$$\text{and} \quad 1 \le \upsilon < 2/(2-\beta), \tag{4.2}$$

where $p \equiv 2 + \Gamma(0)$.

Proof. Let $\hat{g}$ be defined as in (2.17c). From the definition of G in Lemma 2.1, Assumption (A3), and Remark 2.1 we can assume that $\hat{g}(r) \equiv \sum_{i=1}^{\infty} g_i r^i$ near $r = 0$ with $|g_i| \le \eta^i$ for all $i \ge 1$ with $\eta > 0$ as small as we please. Hence we can apply Corollary 3.1 to the system (2.17), yielding that the general solution has the form (3.2a) about $r = 0$ if $k \ne -n$. We note that in this case one can obtain $X_{-n}(r)$ directly from $X_n(r)$. For all integers k the constant C_k^- in (3.2a) must be set equal to zero since $\lambda_k^- < 0$, $\hat{a}_k^{\pm}(0) + \hat{b}_k^{\pm}(0) \equiv 1$ and we know that $X_k \in C[0,\delta]^2$; see (2.15) and (2.17). For the same reason $C_k^+ \equiv 0$ when $|k| < n$ since $\lambda_k^+ < 0$. In addition, as $H(0) = 0$, $\lambda_n^+ = 0$, $\hat{a}_n^+(0) = 1$, and $\hat{b}_n^+(0) = 0$ it follows that $C_n^+ = 0$. Therefore we

Downloaded 09/18/13 to 128.101.152.17. Redistribution subject to SIAM license or copyright; see http://www.siam.org/journals/ojsa.php

have from (3.6) and (2.17) that

$$H(r,\theta) \equiv \sum_{k\geq|n+1|} C_k^+ r^{\lambda_k^+}(\hat{a}_k^+(r)+\hat{b}_k^+(r))e^{i(k-n)\theta}$$

$$\equiv \sum_{k=n+1}^{\infty}\Big[\Big(C_k^+e^{ik\theta}+\varepsilon_k^+\overline{C_k^+}e^{-ik\theta}\Big)\hat{a}_k^+(r) \tag{4.3a}$$

$$+(C_k^+e^{ik\theta}+\varepsilon_k^-\overline{C_k^+}e^{-ik\theta})\hat{b}_k^+(r)\Big]r^{\lambda_k^+}e^{-in\theta}$$

or equivalently,

$$H(\xi) \equiv \sum_{k=n+1}^{\infty}\Big[\Big(C_k^+\xi^k+\varepsilon_k^+\overline{C_k^+}\bar{\xi}^k\Big)\hat{a}_k^+(|\xi|) \tag{4.3b}$$

$$+\Big(C_k^+\xi^k+\varepsilon_k^-\overline{C_k^+}\bar{\xi}^k\Big)\hat{b}_k^+(|\xi|)\Big]|\xi|^{\lambda_k^++n-k}\xi^{-n}.$$

Since $H\in L^2(B)$ we have from (4.3) that for R sufficiently small,

$$\int_0^R r\int_0^{2\pi}|H(r,\theta)|^2d\theta dr$$

$$\equiv 2\pi\sum_{k=n+1}^{\infty}|C_k^+|^2\int_0^R r^{2\lambda_k^++1}[(\hat{a}_k^+(r)+\hat{b}_k^+(r))^2 \tag{4.4}$$

$$+(\varepsilon_k^+\hat{a}_k^+(r)+\varepsilon_k^-\hat{b}_k^+(r))^2]dr<\infty.$$

In the analysis that follows M_i denotes a constant dependent only on $R<1$ and p. We set $\sigma_i\equiv a_i^+(k)+b_i^+(k)$, $t=1/r$ and note from (3.2b) that $\lambda_k^+\leq k(p-1)^{1/2}$. In addition we note that the L^1 norm over an interval of length 2 of a polynomial of degree k with leading coefficient 1 is bounded below by 2^{1-k}; see, for example, p. 80 in [25]. Then for $k\geq n+1$ we have from (3.8) that

$$R\left[\int_0^R r^{2l_k^++1}[\hat{a}_k^+(r)+\hat{b}_k^+(r)]^2dr\right]^{1/2}$$

$$\geq\int_0^R r^{\lambda_k^++1}|\hat{a}_k^+(r)+\hat{b}_k^+(r)|dr$$

$$\geq\int_0^R r^{\lambda_k^++1}\left[\left|\sum_{i=0}^{k}\sigma_i r^i\right|-2[(8\mathcal{D}\eta k)^k/k!]\sum_{i=k+1}^{\infty}r^i\right]dr \tag{4.5}$$

$$\geq\int_{1/R}^{\infty}t^{-(\lambda_k^++k+3)}\left|\sum_{i=0}^{k}\sigma_i t^{k-i}\right|dt-(M_1\eta k)^k/k!$$

$$\geq M_2^k \int_{1/R}^{(2R+1)/R} \left| \sum_{i=0}^{k} \sigma_i t^{k-i} \right| dt - (M_1 \eta k)^k / k!$$

$$\geq M_3^k - (M_1 \eta k)^k / k! \geq M_3^k - (M_1 \eta e)^k,$$

since $\sigma_1 \equiv 1$ and $(k/e)^k/k! \leq e^{-1}$. Combining (4.4) and (4.5) we have that by choosing η sufficiently small there exists an M, dependent only on R and p, such that

$$|C_k^+| \leq M^k \quad \text{for all } k \geq n+1. \tag{4.6}$$

Let δ be such that δ, $Me\delta^{(p-1)^{1/2}} \in (0, R)$. As $\lambda_k^+ \geq k(p-1)^{1/2} - pn/2$ we have that $\sum_{k=n+1}^{\infty} (Me)^k \delta^{\lambda_k^+} < \infty$. It then follows from (4.3b), (4.6), (3.2d), and (3.8) that for $|\xi| \leq \delta$,

$$\begin{aligned} |H(\xi)| &\leq \sum_{k=n+1}^{\infty} |C_k^+| \left[(1 + |\varepsilon_k^+|) |\hat{a}_k^+(|\xi|)| + (1 + |\varepsilon_k^-|) |\hat{b}_k^+(|\xi|)| \right] |\xi|^{\lambda_k^+} \\ &\leq C(|\xi|/\delta)^{\lambda_{n+1}^+} \sum_{k=n+1}^{\infty} [|\hat{a}_k^+(|\xi|)| + |\hat{b}_k^+(|\xi|)|] M^k \delta^{\lambda_k^+} \qquad (4.7) \\ &\leq C|\xi|^{\lambda_{n+1}^+} \sum_{k=n+1}^{\infty} (Me)^k \delta^{\lambda_k^+} \leq C|\xi|^{\lambda_{n+1}^+}. \end{aligned}$$

Therefore the upper bound of (41) in [13] holds. In addition, we see from (4.7) that the series (4.3b) is uniformly and absolutely convergent for $|\xi| \leq \delta$.

As H is a one-to-one mapping near $\xi = 0$ it follows that $|C_{n+1}^+| \neq 0$. Noting from (3.2d) that $|\varepsilon_{n+1}^+| < 1$, it follows from (4.3b), (4.7), and (3.8) that there exist positive constants M_i such that for $|\xi|$ sufficiently small,

$$\begin{aligned} |H(\xi)| &\geq |C_{n+1}^+| |\xi|^{\lambda_{n+1}^+} \left[M_1 |\hat{a}_{n+1}^+(|\xi|)| - M_2 |\hat{b}_{n+1}^+(|\xi|)| \right] - M_3 |\xi|^{\lambda_{n+2}^+} \\ &\geq |C_{n+1}^+| M_1 |\xi|^{\lambda_{n+1}^+} \left[1 - M_4 |\xi| - M_5 |\xi|^{\lambda_{n+2}^+ - \lambda_{n+1}^+} \right] \geq |C_{n+1}^+| M_6 |\xi|^{\lambda_{n+1}^+}. \end{aligned}$$

Hence, the lower bound of (41) in [13] holds. Similarly, one can prove that the bounds (42) and (43) in [13] hold. The remainder of the proof of (4.1) is then identical to that in [13]; that is, apply Lemmas 1 and 2 and Corollary 2 in [13]. □

Note that the conclusion of Theorem 4.1 is not necessarily true for (1.1) for $\rho \neq 0$ (even if ρ is a constant); see [29].

Example 4.1. For the examples considered in Example 2.1 we have with $q > 1$ that the local weak solutions of (2.3) are such that (4.1) and (4.2) hold with $\rho \equiv 2 + \Gamma(0)$ given as follows: (i) $\varphi(t) \equiv t^{q-2} \Rightarrow p = q$; (ii) $\varphi(t) \equiv [t(1+t)]^{(q-2)/2} \Rightarrow p = (q+2)/2$; and (iii) $\varphi(t) \equiv [t(1+t^2)]^{(q-2)/3} \Rightarrow p = (q+4)/3$. Therefore we see that the solutions of (2.3) with φ defined by (ii) and (iii) are as regular as those of (2.3) with $\varphi(t) \equiv t^{p-2}$, which is nothing other than the singular part of φ in (ii) and (iii). In other words, it is the singular part of φ that determines the regularity of the local solutions of (2.3).

Downloaded 09/18/13 to 128.101.152.17. Redistribution subject to SIAM license or copyright; see http://www.siam.org/journals/ojsa.php

Downloaded 09/18/13 to 128.101.152.17. Redistribution subject to SIAM license or copyright; see http://www.siam.org/journals/ojsa.php

5. A global regularity result. As mentioned in §1, in order to obtain explicit error bounds for the finite element approximation of (2.3) it is necessary to extend the local regularity results obtained in the last section into global results. Apart from those global $C^{1,\alpha}$ and H^2 results mentioned in the introduction, there appear to be no other global results in the literature. We derive in this section a global regularity result from Theorem 4.1 under some restrictions on the boundary data.

Consider the following mechanics problem. Suppose a plane Ω, which is made of a power law material, is undergoing longitudinal shear (the mode III problem; see [3]). In this case the only nonvanishing component of displacement is the z-component $u(x,y)$, which satisfies (2.3) with $\varphi(t) \equiv t^{p-2}$ and $p \in (1,2)$. Physically one expects that the tangential strain of the plane along $\partial\Omega$ will vary strictly monotonically except at those points where the displacement $u(x,y)$ achieves its extreme values. Motivated by this observation we will suppose that (2.3) is supplemented by the boundary condition

$$u|_{\partial\Omega} = g|_{\partial\Omega}, \tag{5.1}$$

where g has a nonvanishing tangential derivative, g_τ, on $\partial\Omega$ except at those points where g achieves its global extrema on $\partial\Omega$. We have the following result.

THEOREM 5.1. *Let φ satisfy* (A1) *with $C^\infty(0,\infty)$ replaced by $C^{1,1}(0,\infty)$,* (A2) *and* (A3) *with $p \equiv 2 + \Gamma(0) \in (1,2)$. Let $\partial\Omega \in C^{2,1}$ and $g \in C^1(\bar{\Omega})$ be such that $|g_\tau(x,y)| > 0$ if $(x,y) \in \partial\Omega$ and $g(x,y)$ is not a global extremum of g on $\partial\Omega$. Let $u \in W^{1,q}(\Omega)$ be the unique solution of* (2.3) *with the boundary condition* (5.1). *Then*

(i) $g \in C^{2,\gamma}(\bar{\Omega}), \qquad \gamma > 0, \Rightarrow u \in C^{2,\mu}(\bar{\Omega}),$

(ii) $g \in C^{2,\gamma}(\bar{\Omega}) \cap W^{3,r}(\Omega), \qquad \gamma > 0$ *and* $r \in (1,p], \Rightarrow u \in C^{2,\mu}(\bar{\Omega}) \cap W^{3,r}(\Omega),$

where $\mu = \gamma$ for $p \in (1, 1.3), \mu = \min(\gamma, \beta)$ for $p \in [1.3, 2)$, and

$$\beta = \{-5 + 1/(p-1) + [1 + 14/(p-1) + 1/(p-1)^2]^{1/2}\}/6. \tag{5.2}$$

Proof. From [19] we have that $g \in C^{1,\alpha}(\bar{\Omega}) \Rightarrow u \in C^{1,\alpha}(\bar{\Omega})$ for some $\alpha > 0$, so that $|\nabla u|$ is continuous on $\bar{\Omega}$. We first show that $|\nabla u|_{\partial\Omega}| \geq \delta > 0$. If $(x,y) \in \partial\Omega$ and $g_\tau(x,y) \neq 0$, then $|\nabla u(x,y)| > 0$ as $(u-g)|_{\partial\Omega} = 0$. Let $g_\tau(x_0,y_0) = 0$ and (without loss of generality) let $g(x,y) \geq g(x_0,y_0)$ for all $(x,y) \in \partial\Omega$. Let $u_0 \equiv g(x_0,y_0)$. Then $u - u_0$ is such that $\nabla.(\varphi(|\nabla(u-u_0)|)\nabla(u-u_0)) \equiv \nabla.(\varphi(|\nabla u|)\nabla u) = 0$ in Ω and $(u-u_0)_{\partial\Omega} \geq 0$ because u_0 is a constant. If u is constant, then the desired regularity result holds. Therefore we can assume that u is not constant. From the strong maximum principle in §3 of [28], it follows that $u > u_0$ in Ω. (This result also holds from the proof of the strong comparison principle in [22].) From the Hopf maximum principle, see [28] and Lemma 3.4 in [12], we have that $|\partial u/\partial n(x_0,y_0)| > 0$, where $\underline{n}$ is the outward unit normal to $\partial\Omega$. Consequently, $|\nabla u|_{\partial\Omega}| > 0$ and from the continuity of $|\nabla u|$, there exists a domain $S \subset\subset \Omega$ such that $\partial S \in C^{2,1}$ and $|\nabla u|_{\bar{\Omega}\setminus S}| \geq \delta > 0$.

Downloaded 09/18/13 to 128.101.152.17. Redistribution subject to SIAM license or copyright; see http://www.siam.org/journals/ojsa.php

Let $S \subset\subset S_1 \subset\subset \Omega$. Then from Theorem 4.1 we have that

$$p \in (1, 1.3) \Rightarrow k \geq 3 \Rightarrow u \in C^3(\overline{S_1}), \tag{5.3a}$$

$$p \in [1.3, 2) \Rightarrow k = 2 \Rightarrow u \in C^{2,\beta}(\overline{S_1}) \cap W^{3,p}(S_1) \quad \text{with } \beta \text{given by (5.2).} \tag{5.3b}$$

Rewriting (2.3), we have for $Q \equiv \Omega \backslash S$ and $g \in C^{2,\gamma}(\bar{\Omega})$ that

$$\Delta u + \Gamma(|\nabla u|)\left[(u_x)^2 u_{xx} + 2u_x u_y u_{xy} + (u_y)^2 u_{yy}\right] / |\nabla u|^2 = 0 \quad \text{in } Q,$$

$$u|_{\partial Q} \in C^{2,\mu}(\partial Q).$$

As $\Gamma(t) \equiv [(\varphi(t)t)'/\phi(t)] - 1 \geq q - 2$, it follows from our assumptions that $\Gamma \in C^{0,1}(0, \infty)$ and hence the coefficients $\Gamma(|\nabla u|)(u_x)^2/|\nabla u|^2$, $\Gamma(|\nabla u|)(u_x u_y)/|\nabla u|^2$, and $\Gamma(|\nabla u|)(u_y)^2/|\nabla u|^2 \in C^{0,\alpha}(\bar{Q})$. It then follows from Theorem 9.19 in [12], for example, that $u \in C^{2,\sigma}(\bar{Q})$, $\sigma = \min(\alpha, \mu)$. This yields that the above coefficients then belong to $C^{0,1}(\bar{Q})$ and hence $u \in C^{2,\mu}(\bar{Q})$. Combining this with (5.3) yields that $u \in C^{2,\mu}(\bar{\Omega})$. If $g \in C^{2,\alpha}(\bar{\Omega}) \cap W^{3,r}(\Omega)$, $r \in (1, p]$ it follows in a similar manner that $u \in C^{2,\alpha}(\bar{\Omega}) \cap W^{3,r}(\Omega)$. □

It is of great interest to know if the conclusion of Theorem 5.1 is true or not for more general bounary conditions. Of course, it would be even better if Theorem 4.1 could be extended to a global result. Nevertheless, the above result shows that the global regularity on u required to guarantee the optimal rate of convergence in $W^{1,p}(\Omega)$ of the continuous piecewise linear finite element approximation of (2.3) does hold for suitable data. Another unsolved problem is extending the sharp local regularity results in [13] to higher dimensions.

REFERENCES

[1] G. Aronsson, *Representation of a p-harmonic function near a critical point in the plane*, Manuscripta Math., 66 (1989), pp. 73–95.

[2] G. Aronsson and P. Lindqvist, *On p-harmonic functions in the plane and their stream functions*, J. Differential Equations, 74 (1988), pp. 157–178.

[3] C. Atkinson and C. R. Champion, *Some boundary value problems for the equation* $\nabla \cdot(|\nabla\phi|^N \nabla\phi) = 0$, Quart. J. Mech. Appl. Math., 37 (1984), pp. 401–419.

[4] C. Atkninson and C. W. Jones, *Similarity solutions in some non-linear diffusion problems and in boundary-layer flow of a pseudo plastic fluid*, Quart. J. Mech. Appl. Math., 27 (1974), pp. 193–211.

[5] J. Baranger and K. Najib, *Numerical analysis for quasi-Newtonian flow obeying the power or the Carreau law*, Numer. Math., 58 (1990), pp. 35–49.

[6] J. W. Barrett and W. B. Liu, *Finite element approximation of the p-Laplacian*, Math. Comp., to appear.

[7] B. Bojarski and T. Iwaniec, *p-harmonic equation and quasi regular mappings*, in Partial Differential Equations, Banach Center Publications 19, PWN-Polish Scientific Publishers, Warsaw, 1987, pp. 25–38.

[8] S. S. Chow, *Finite element error estimates for non-linear elliptic equations of monotone type*, Numer. Math., 54 (1989), pp. 373–393.

[9] E. DiBenedetto, *$C^{1+\alpha}$ local regularity of weak solutions of degenerate elliptic equations*, Nonlinear Anal., 7 (1983), pp. 827–850.

[10] E. DiBenedetto and A. Friedman, *Regularity of solutions of nonlinear degenerate parabolic systems*, J. Reine Angew. Math., 349 (1984), pp. 83–128.

[11] M. Giaquinta, *Multiple Integrals in the Calculus of Variations and Nonlinear Elliptic Systems*, Princeton Univ. Press, Princeton, NJ, 1983.

[12] D. GILBARG AND N. TRUDINGER, *Elliptic Partial Differential Equations of Second Order*, Second ed., Springer-Verlag, Berlin, 1984.
[13] T. IWANIEC AND J. J. MANFREDI, *Regularity of p-harmonic functions on the plane*, Rev. Math. Iberoamericana, 5 (1989), pp. 1–19.
[14] A. KUFNER, O. JOHN, AND S. FUČIK, *Function Spaces*, Nordhoff, Leyden, 1977.
[15] O. A. LADYZHENSKAYA AND N. N. URAL'TSEVA, *Linear and Quasilinear Elliptic Equations*, Academic Press, New York, 1968.
[16] O. LEHTO AND K. VIRTANEN, *Quasiconformal Mappings in the Plane*, Springer-Verlag, Berlin, 1973.
[17] J. L. LEWIS, *Smoothness of certain degenerate elliptic equations*, Proc. Amer. Math. Soc., 80 (1980), pp. 259–265.
[18] ———, *Regularity of the derivatives of solutions to certain elliptic equations*, Indiana Univ. Math. J., 32 (1983), pp. 849–858.
[19] G. M. LIEBERMAN, *Boundary regularity for solutions of degenerate elliptic equations*, Nonlinear Anal., 12 (1988), pp. 1203–1219.
[20] W. B. LIU AND J. W. BARRETT, *A remark on the regularity of the solutions of the p-Laplacian and its applications to their finite element approximation*, J. Math. Anal. Appl., to appear.
[21] ———, *A further remark on the regularity of the solutions of the p-Laplacian and its applications to their finite element approximation*, Nonlinear Anal., to appear.
[22] J. J. MANFREDI, *p-harmonic functions in the plane*, Proc. Amer. Math. Soc., 103 (1988), pp. 473–479.
[23] C. B. MORREY, *Multiple Integrals in the Calculus of Variations*, Springer-Verlag, Berlin, 1966.
[24] J. R. PHILIP, *N-diffusion*, Austral. J. Phys., 14 (1961), pp. 1–13.
[25] T. J. RIVLIN, *The Chebyshev Polynomials*, John Wiley, New York, 1974.
[26] F. DE THELIN, *Local regularity properties for the solutions of a nonlinear partial differential equation*, Nonlinear Anal., 6 (1982), pp. 839–844.
[27] P. TOLKSDORF, *On quasilinear boundary value problems in domains with corners*, Nonlinear Anal., 5(1981), pp. 721–735.
[28] ———, *On the Dirichlet problem for quasilinear equations in domains with conical boundary points*, Comm. Differential Eqns., 8 (1983), pp. 773–817.
[29] ———, *Regularity for a more general class of quasilinear elliptic equations*, J. Differential Eqns., 51 (1984), pp. 126–150.
[30] K. UHLENBECK, *Regularity for a class of nonlinear elliptic systems*, Acta. Math., 138 (1977), pp. 219–240.
[31] N. N. URAL'TSEVA, *Degenerate quasilinear elliptic systems*, Zap. Nauchno. Sem. Leningrad. Otdel. Mat. Inst. Steklov., 7 (1968), pp. 184–222. (In Russian.)

Downloaded 09/18/13 to 128.101.152.17. Redistribution subject to SIAM license or copyright; see http://www.siam.org/journals/ojsa.php